

Instructional Goal

To help 4th–6th grade learners master U.S. state-capital associations and geographic understanding through multimodal, UDL-aligned HTML5 games.

Learning Outcomes

Level 1 – Match & Memorize: Identify and recall the capital of a typed U.S. state.

Level 2 – Geographic Context: Explain a state's location and geographic feature once recognized.

UDL Framework Integration

- Representation: Visual map, text input, audio (TTS), hints
- Action/Expression: Typing state names, receiving feedback, switching game levels
- Engagement: Choice of difficulty, feedback loops, real-time score/progress display

Functional Component Breakdown

- State Input Field – Learner types in the state name
- Capital Display – Auto-displays corresponding capital with TTS (if correct)
- Geographic Feature – In Level 2, additionally displays descriptive feature (mountain, plains...)
- Interactive Map – Static U.S. map with capital labels guides visual attention
- Level Switcher – Tabs/buttons to choose between Level 1 and Level 2
- TTS Feedback – On input or hover, capital/state names spoken aloud
- Progress Tracker – Tracks correct entries for each level

Cognitive Tasks per Level

Level 1 – Input state name → get capital → hear TTS response

Level 2 – Input state name → get capital + geography → hear response → gain point

Planned Enhancements

- Add score system: Each correct match earns 1 point and displays live progress.
- Add local storage: Store learner progress per level to allow resume from last point.

Deployment Context

- Platform: HTML5, fully offline-capable (can be used on LMS, browser, or USB)
- Grade Range: Upper elementary / middle school geography
- Accessibility: Text-to-speech, high contrast, keyboard nav possible